Jenkins CI/CD Pipeline: GitHub Integration, Automated Build, Deployment & Testing on Linux Server

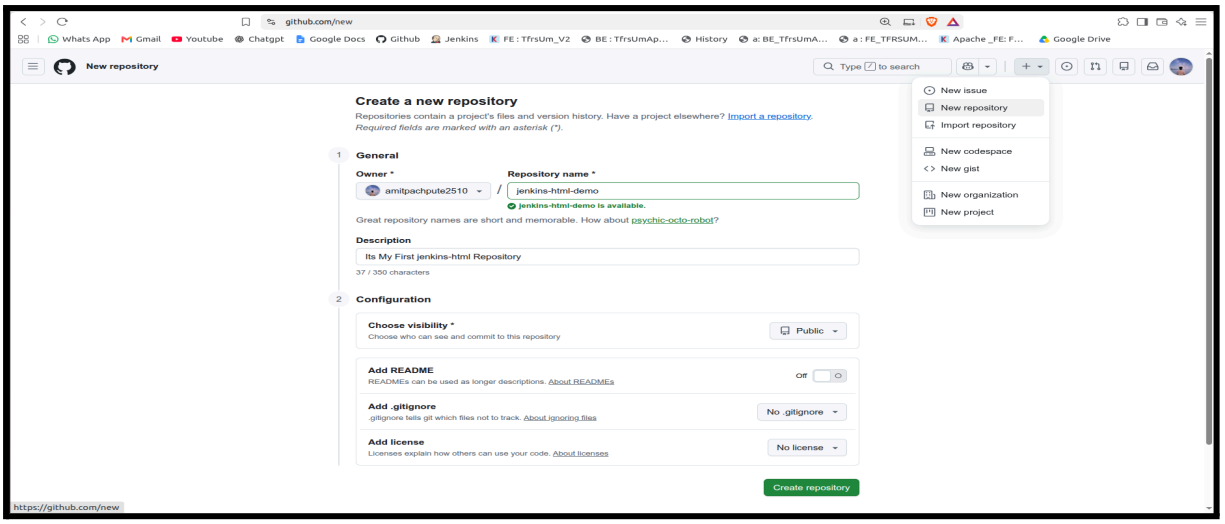
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Step 1: Create a GitHub repository

Go to Your Github Accont and Login
Then Create a GitHub repository

- Repository name: jenkins-html-demo
- Visibility: Public (simplest for learning)
- Add a README: Optional
- Click Create repository.

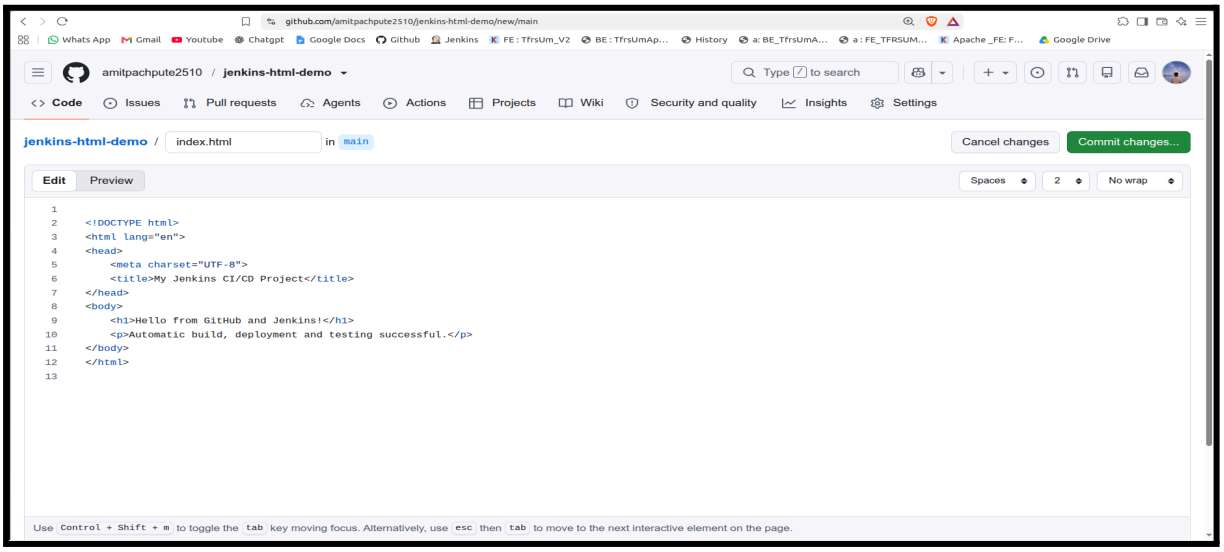
Your project will contain just one website file:



Create index.html on GitHub using Add file → Create new file and paste

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Jenkins CI/CD Project</title>
</head>
<body>
    <h1>Hello from GitHub and Jenkins!</h1>
    <p>Automatic build, deployment and testing successful.</p>
    <h2>My first automatic Jenkins deployment!.</h2>
</body>
```

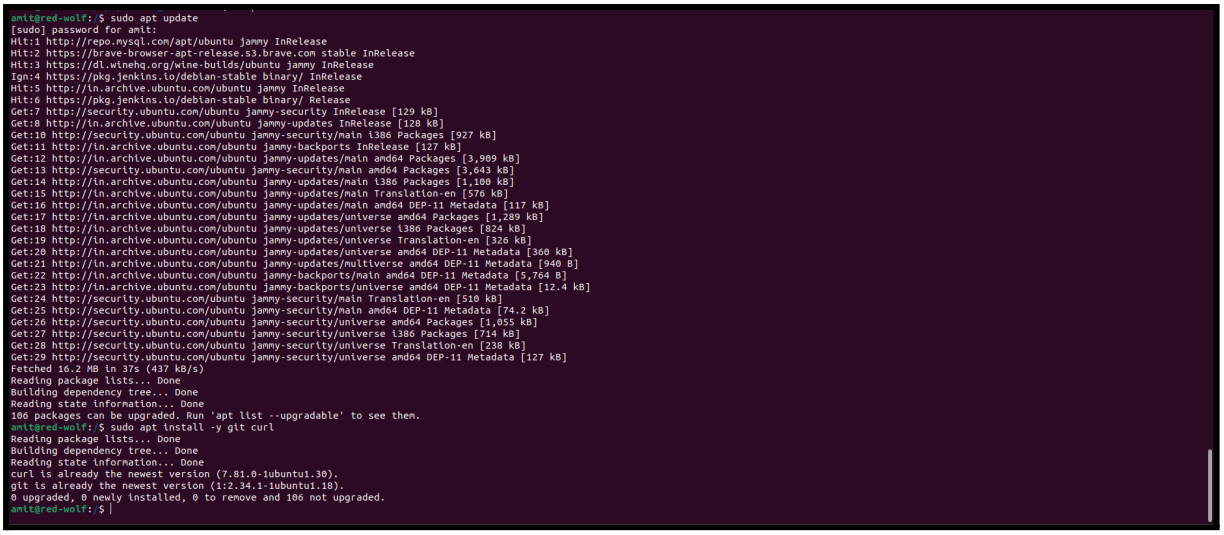
</html>



Step 2: Prepare your Ubuntu server

sudo apt update

sudo apt install -y git curl



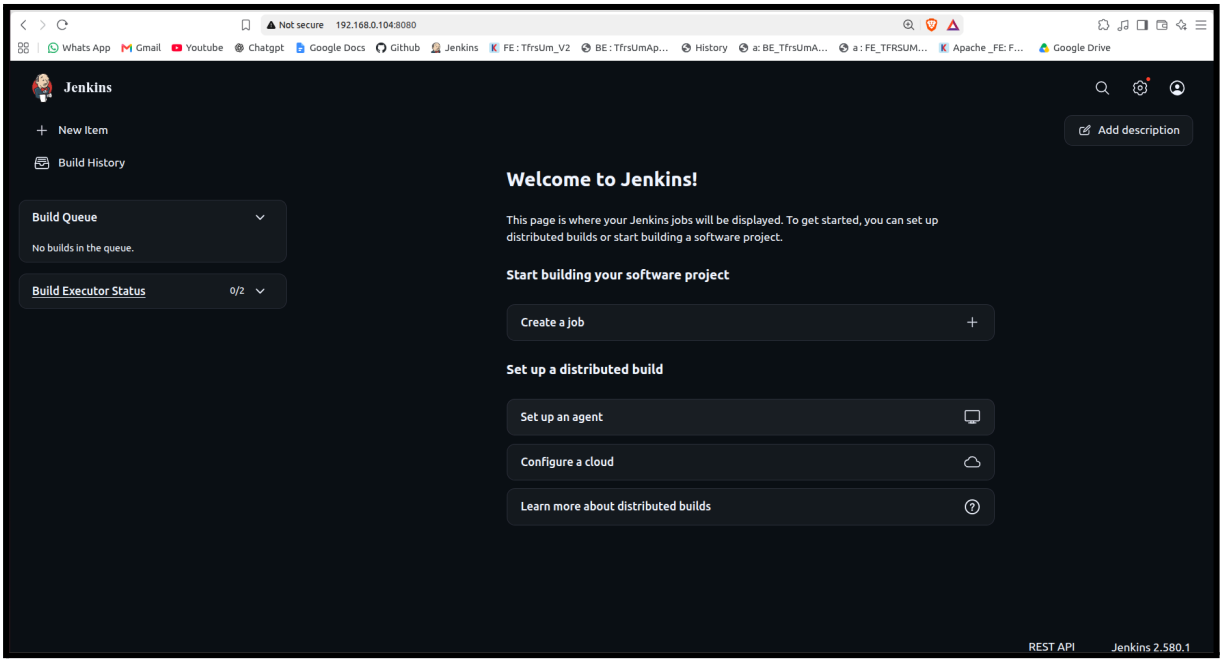
Create a separate deployment directory for this project:

sudo mkdir -p /var/www/html/jenkins-demo

sudo chown jenkins:www-data /var/www/html/jenkins-demo

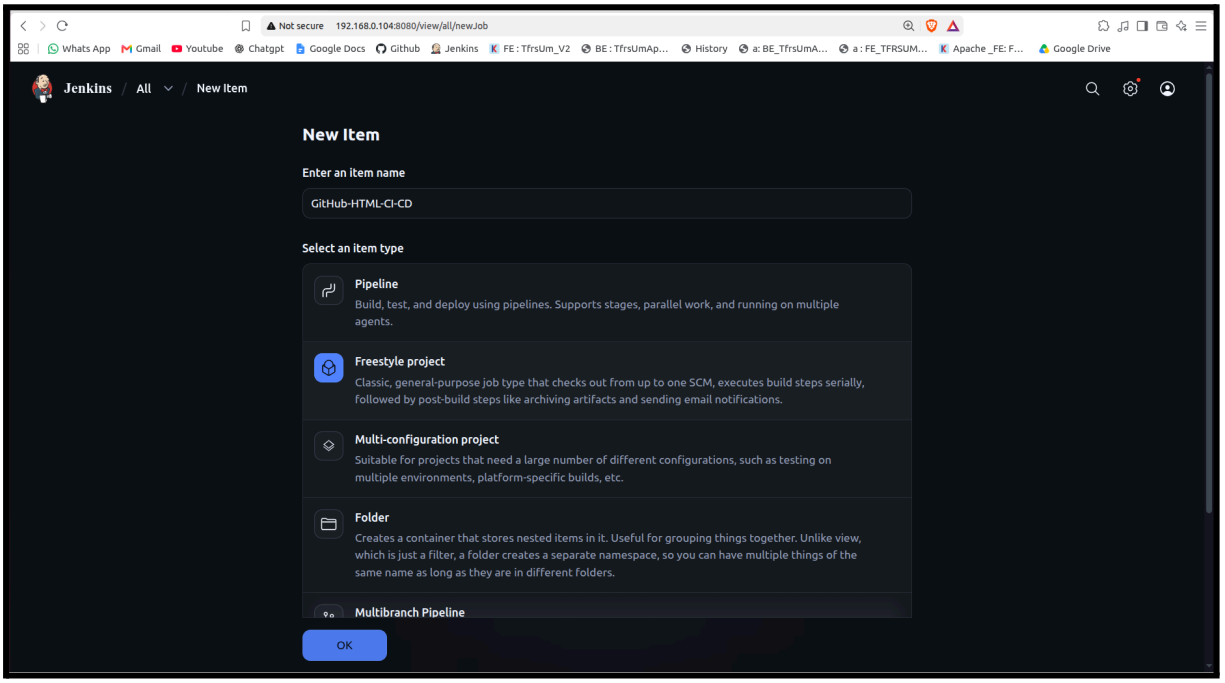
sudo chmod 775 /var/www/html/jenkins-demo

Step 3: Connect GitHub to Jenkins



1. Create a Jenkins job

Click New Item, enter GitHub-HTML-CI-CD, select Freestyle project, and click OK.



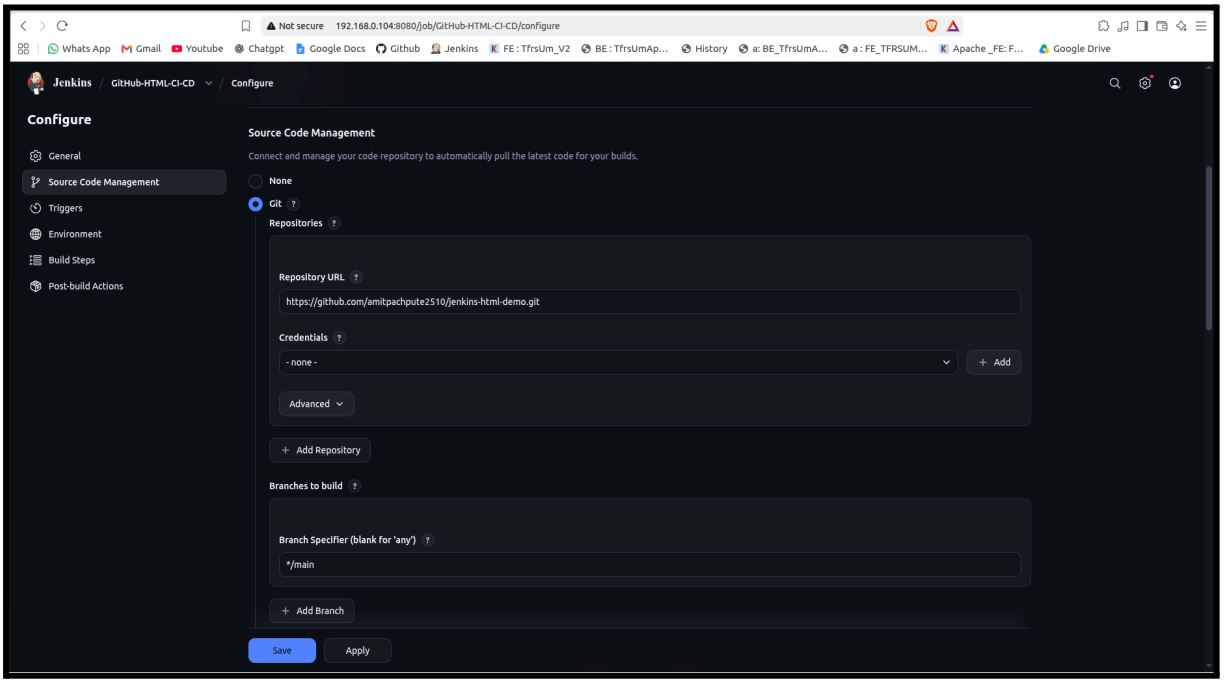
2. Configure Source Code Management

Select Git and enter your repository URL:

https://github.com/YOUR_USERNAME/jenkins-html-demo.git

Replace YOUR_USERNAME with your GitHub username. Set the branch to */main. Leave credentials empty for a public repository.

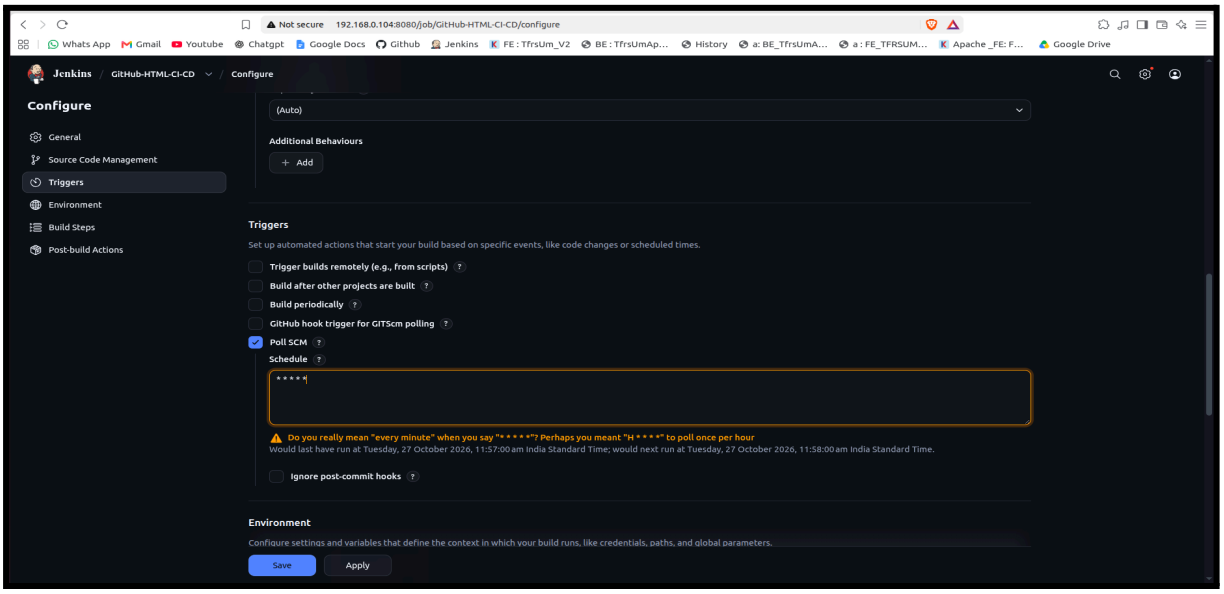
URL : https://github.com/amitpachpute2510/jenkins-html-demo.git



3. Configure automatic polling

Under Build Triggers, select Poll SCM and enter this schedule:

* * * * *



Step 4: Add the build, deployment, and test script

Under Build Steps → Add build step → Execute shell, paste this script:

```
#!/bin/bash
set -eu

SOURCE="$WORKSPACE/index.html"
DEST="/var/www/html/jenkins-demo/index.html"
URL="http://127.0.0.1/jenkins-demo/"

echo "===== BUILD ====="

test -f "$SOURCE"
grep -q "Hello from GitHub and Jenkins!" "$SOURCE"
echo "BUILD SUCCESS"

echo "===== DEPLOY ====="

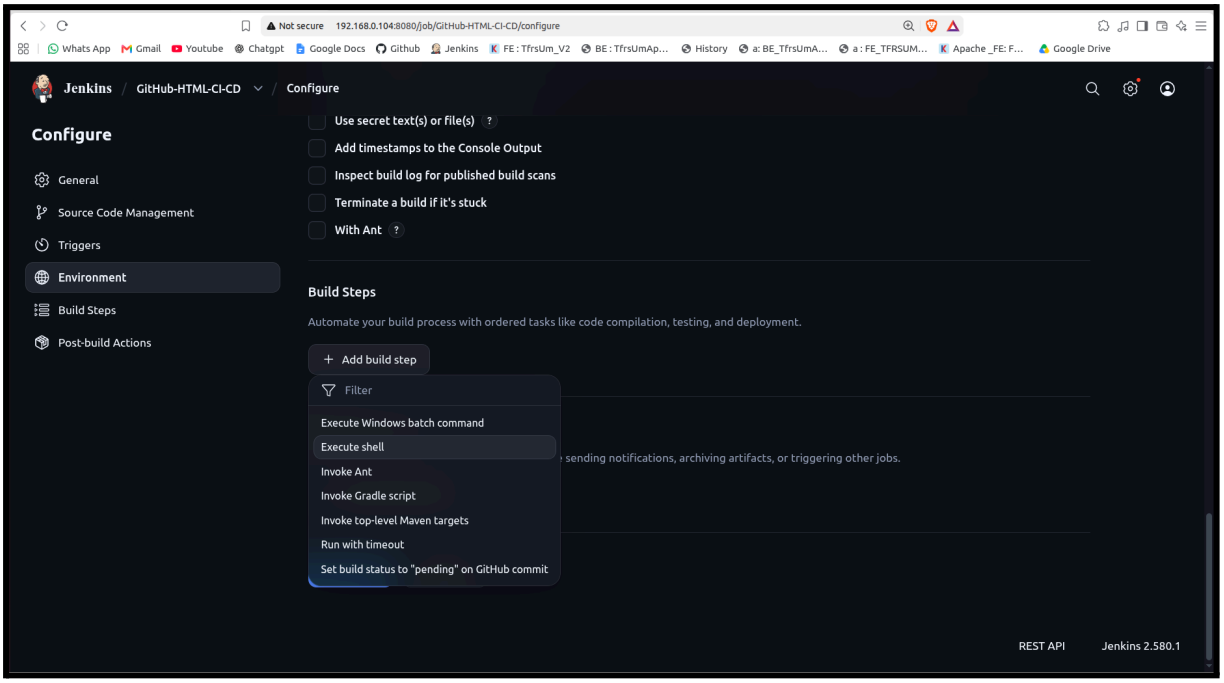
install -m 644 "$SOURCE""DEST"
echo "DEPLOYMENT SUCCESS"

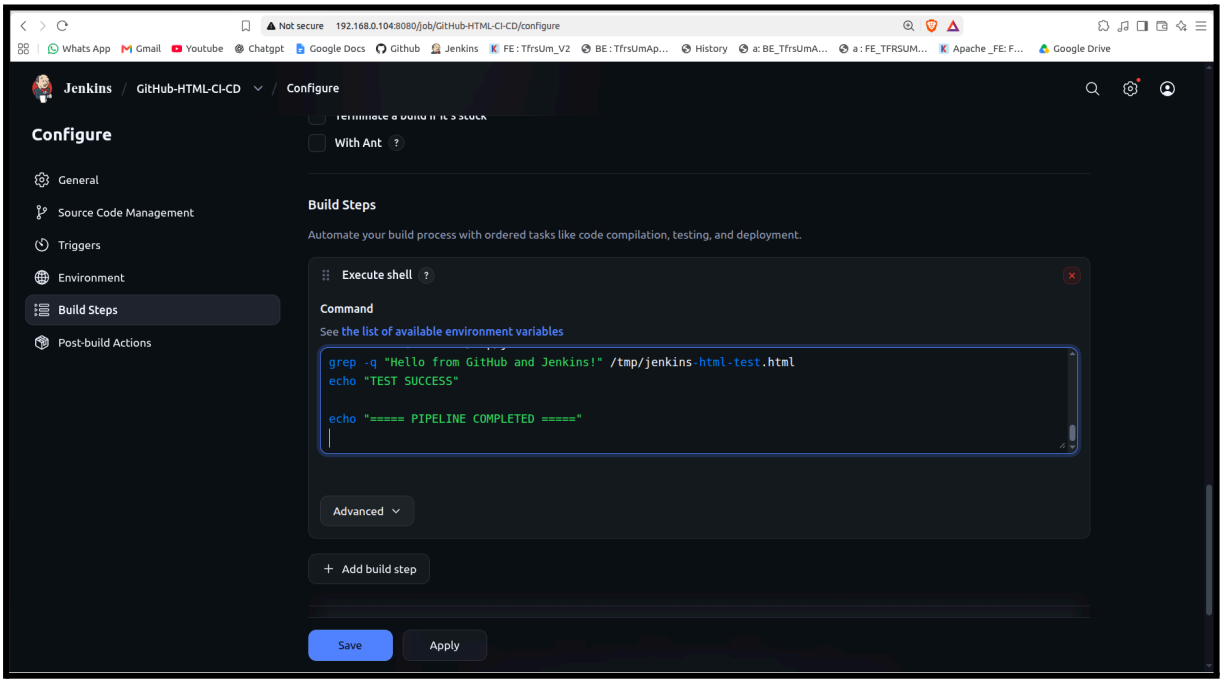
echo "===== TEST ====="

curl -fsS "$URL" -o /tmp/jenkins-html-test.html
grep -q "Hello from GitHub and Jenkins!" /tmp/jenkins-html-test.html
echo "TEST SUCCESS"

echo "===== PIPELINE COMPLETED ====="
```

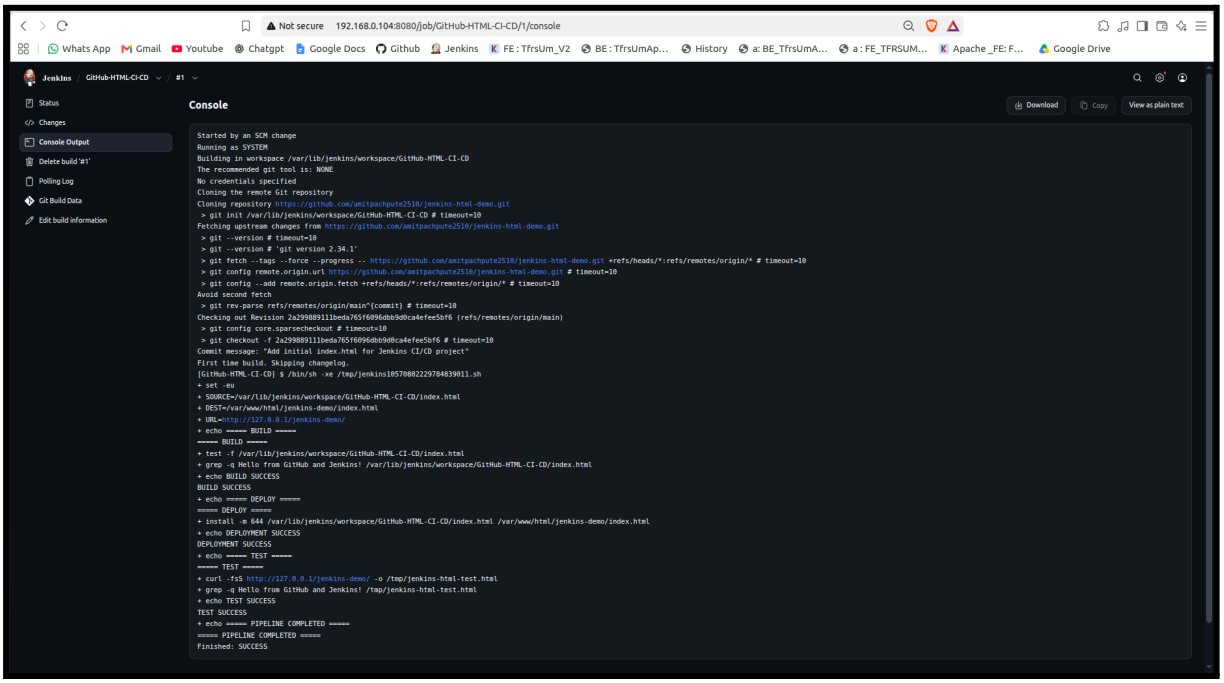
Click **Save**.





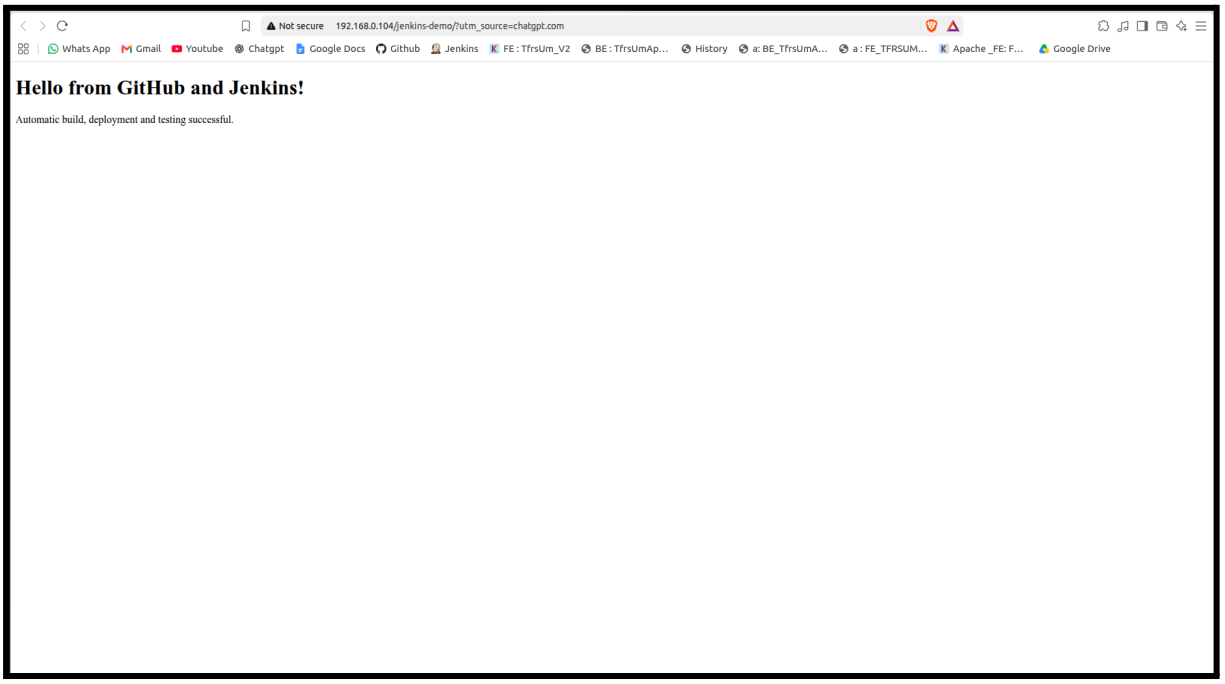
Step 5: Run and verify

1. Click Build Now in your Jenkins job.
2. Open Build History → latest build → Console Output.
3. Confirm the output ends with Finished: SUCCESS.
4. Open your website in a browser.



If your server IP is still 192.168.0.104,

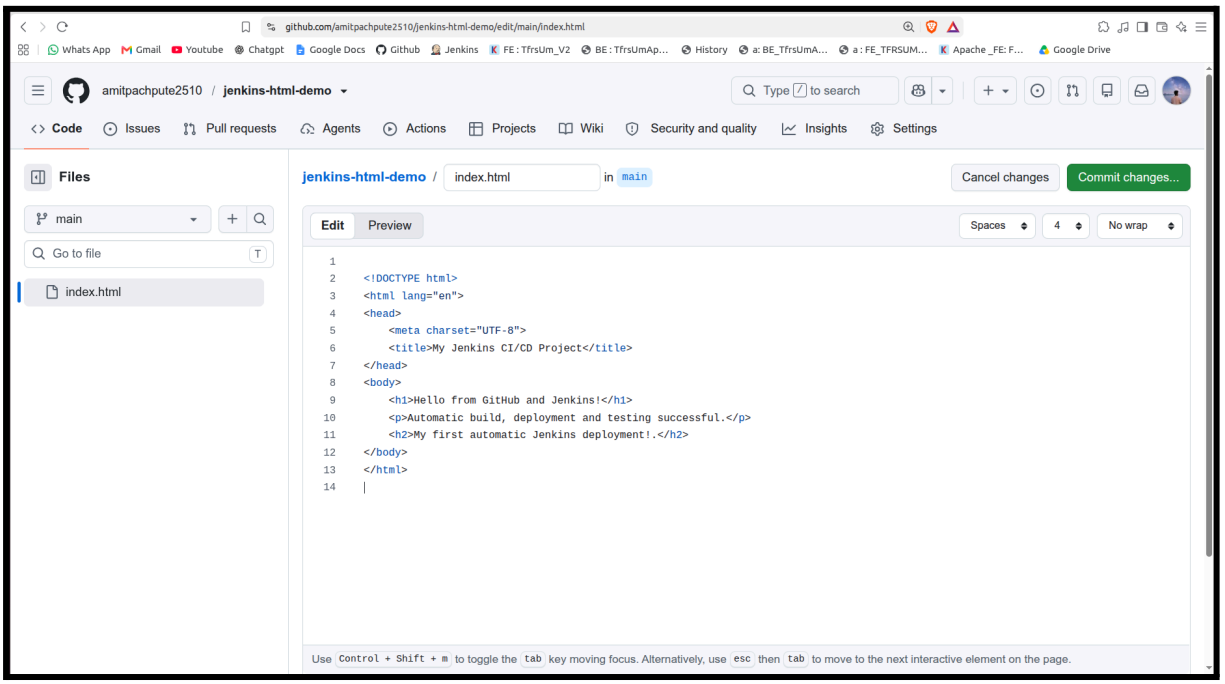
the URL is: http://192.168.0.104/jenkins-demo/



Step 6: Test automatic deployment

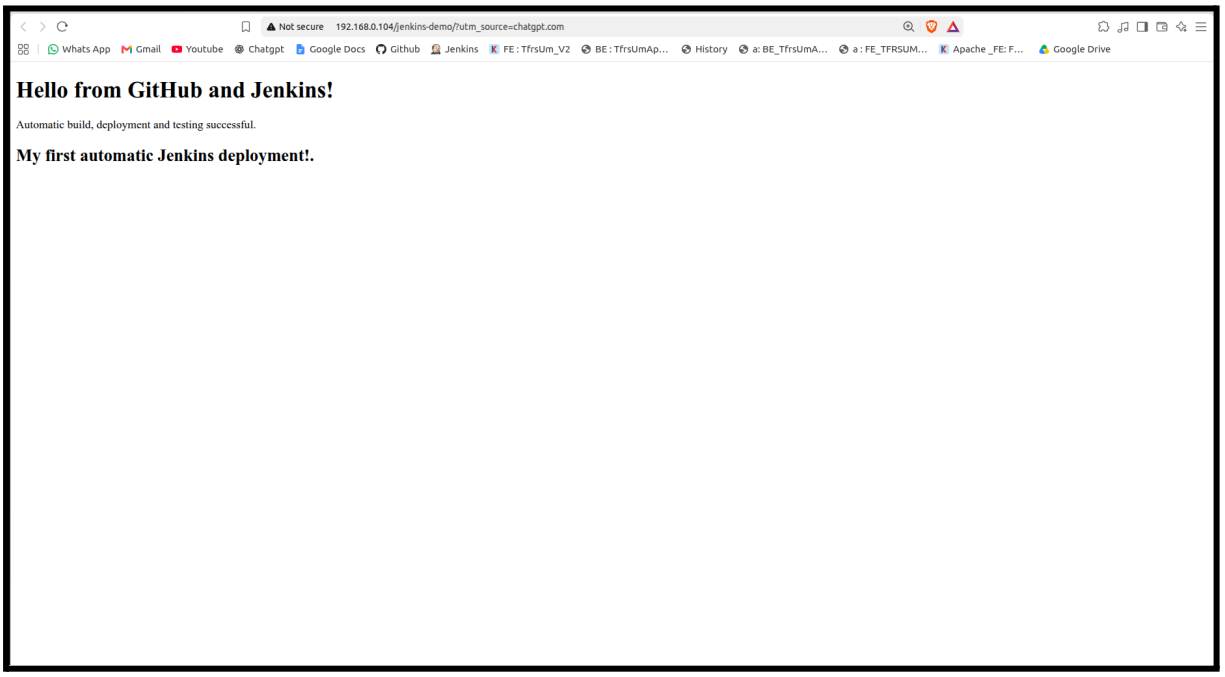
Once the first build succeeds, go back to GitHub and edit index.html. Change the heading to:

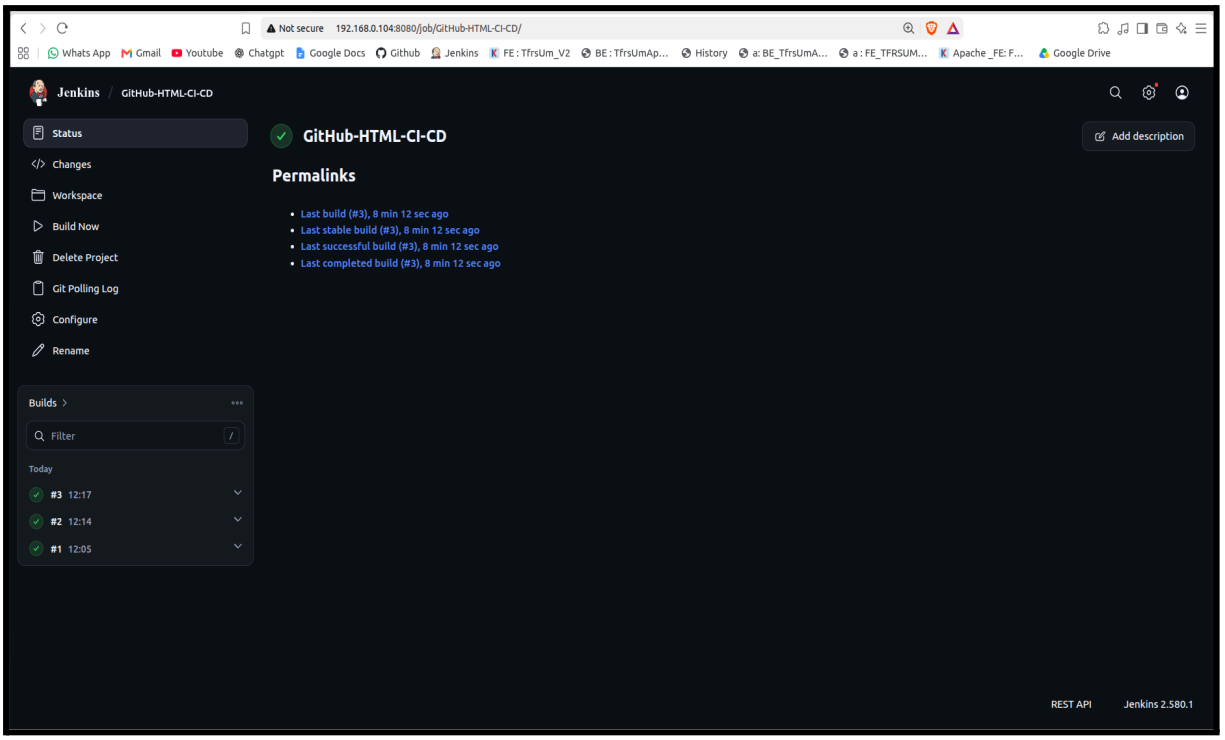
<h1>My first automatic Jenkins deployment!</h1>



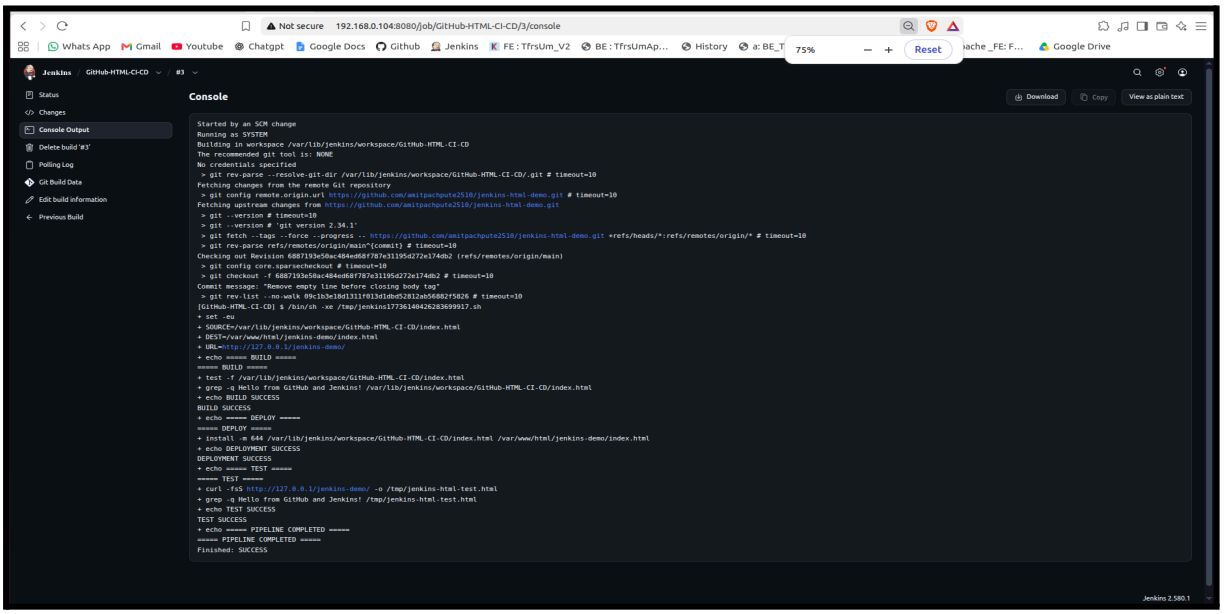
Then click On **Commit the change to main.**

The Changes You are Seen
Refresh Your URL : http://192.168.0.104/jenkins-demo/

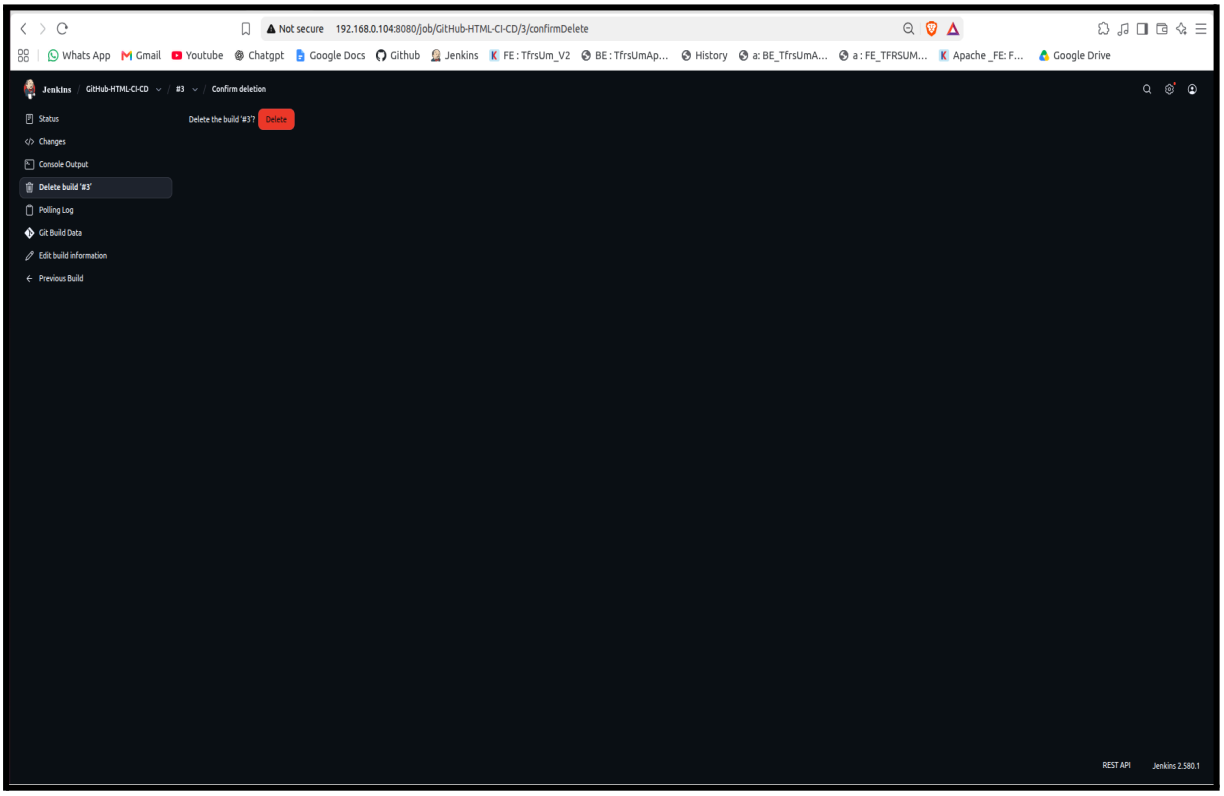




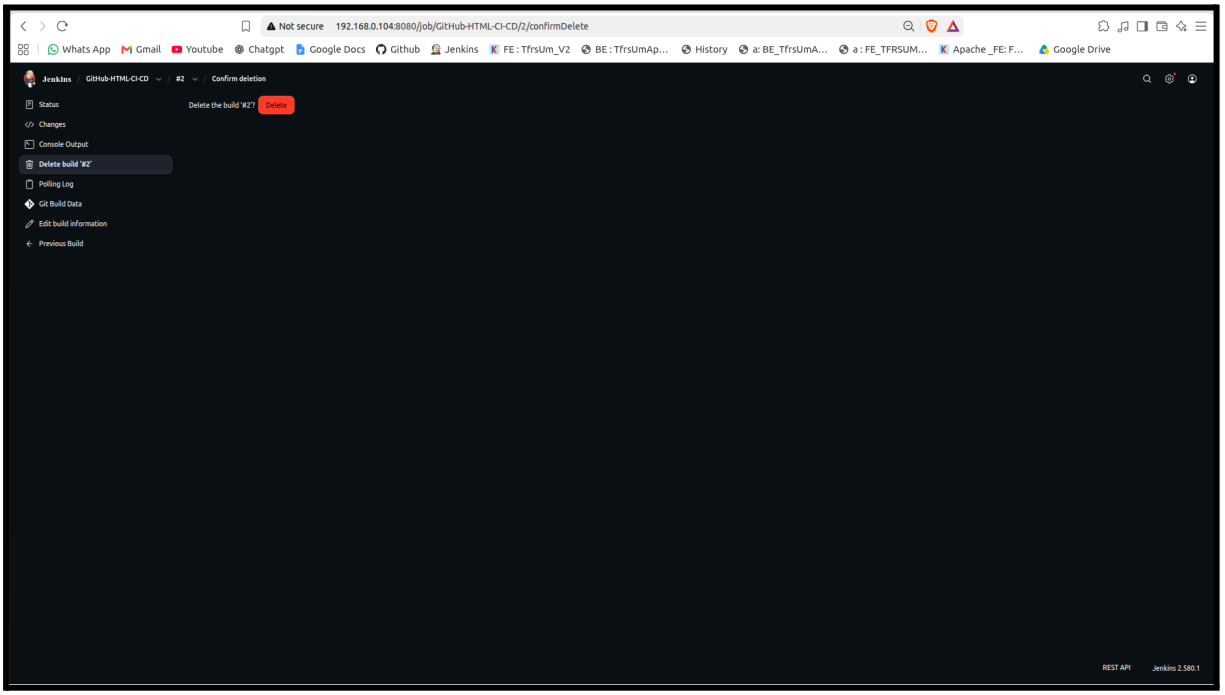
Click On Latest



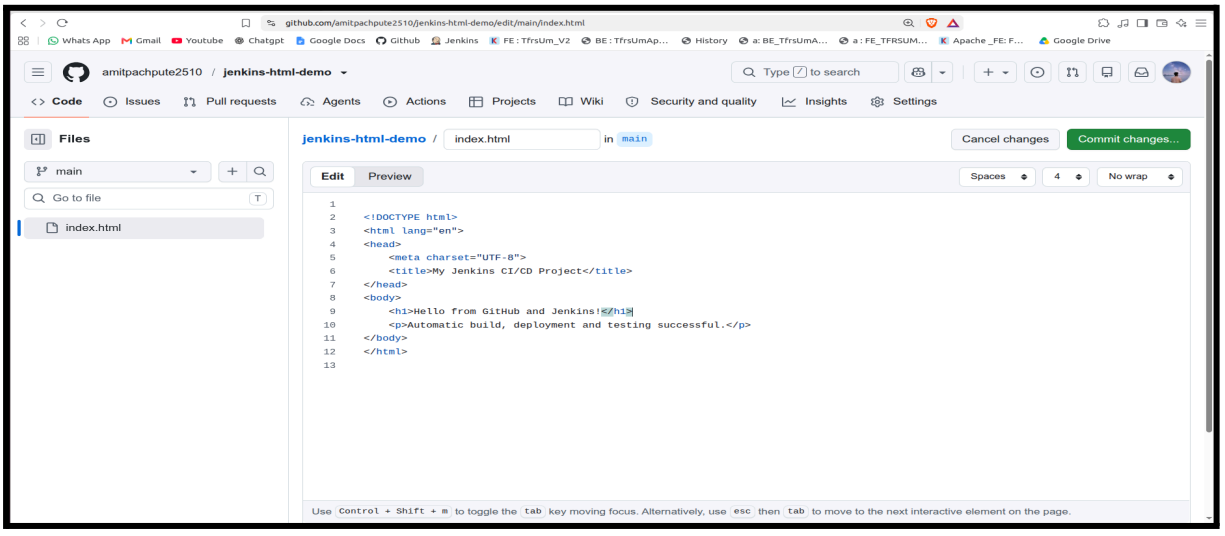
Now Delete The Latest



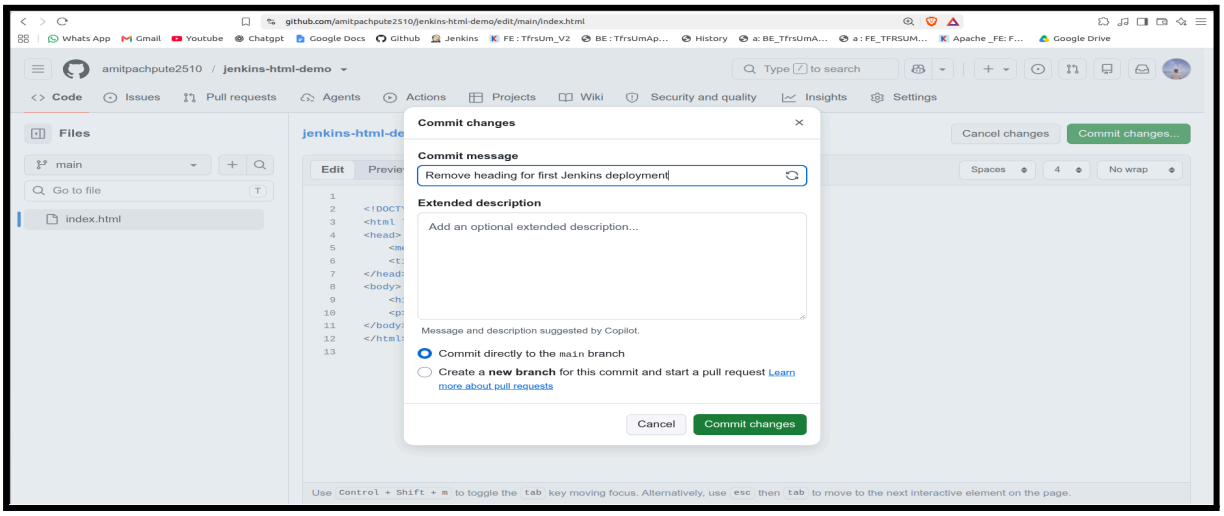
Again Delete the Previously All Build



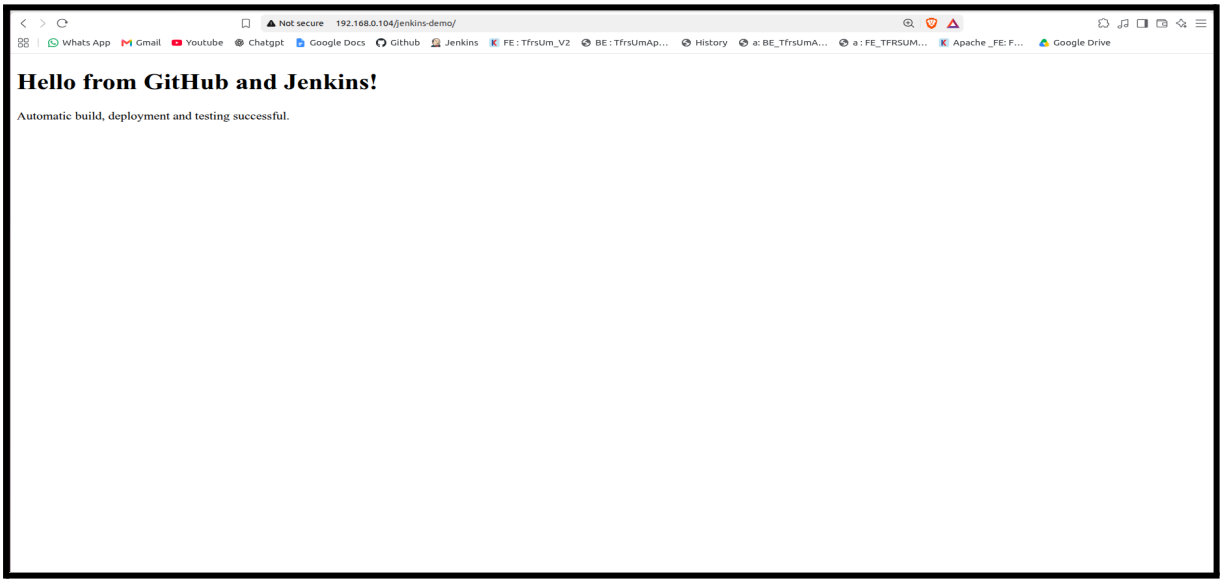
Changes the Index.html Remove that tag Again



Commit changes



Go To URL : http://192.168.0.104/jenkins-demo/



Created By Amit Pachpute